# *Spectral Heart Rate Estimation using AS7265x Sensor*

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Checkpoint 1

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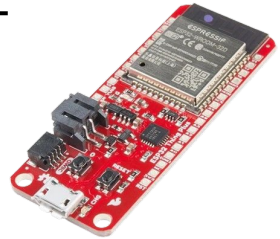
# Our project

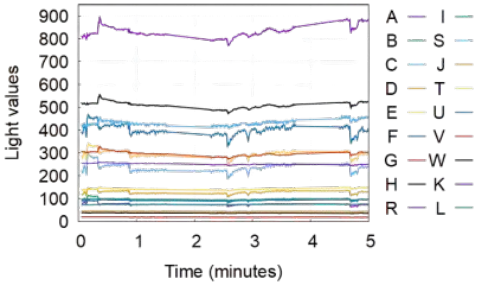
## Goal

To Investigate and develop a multi-spectral optical sensing heart rate sensor using AS7265x.

## Idea

- Measure reflected light from skin at multiple wavelengths
- Observe periodic changes caused by blood flow
- Estimate heart rate from spectral signal variations






# System Overview

**Hardware:**

- ESP32 Microcontroller
- AS7265x Spectral sensor (18 channels: 410-940 nm)

**Ground Truth:**

- Garmin watch broadcasting heart rate via Bluetooth.
- Custom logger application records timestamped HR values

# Current Progress

Completed:

- Implemented BLE heart-rate logger
- Verified live HR capture from Garmin watch
- Logging data to CSV for experiments

Hardware:

- Reviewing spectrometer code, and merging with custom logger application to have both sensory value and ground truth at same time stamp (Idea)
- Sourcing AS7256x module, to have another device for testing (a garmin watch; if possible)

# Next Steps

- Integrate spectrometer data logging
- Collect synchronized spectrometer + HR datasets
- Analyze spectral signals over time
- Estimate heart rate from spectral patterns
- Compare with Garmin ground truth

Code:
https://github.com/mharisirfan/heart-rate-logger

# Thank you!